

Module-IV Question Bank- Server-Side Scripting (PHP)

Part-A (2 mark)

1. Write a Laravel program to create a named route called home that returns the message "Welcome to Amir Raja's PHP World!".
2. What is CSRF protection in Laravel? Why is the @csrf directive used in forms?
3. Write a Laravel POST route for the URL /student that returns Student Form Submitted.
4. What is the purpose of the .env file in Laravel database configuration?
5. Write the Artisan command to create a migration for a students table.
6. What are Laravel Migrations? State any two advantages of using migrations. Write an Eloquent statement to retrieve all records from the Student model. What is Eloquent ORM?
7. Mention any two advantages of using Eloquent.
8. Write an Eloquent query to retrieve employees whose salary is greater than 30000.
9. Differentiate between create() and update() methods in Eloquent.
10. Write an Eloquent statement to delete the employee whose id is 5.
11. What is the purpose of the old() helper function in Laravel forms?
12. Write a Laravel validation statement to validate that name is required and email is a valid email address.
13. What is Flash Data in Laravel? Mention one situation where it is commonly used.
14. Write an Eloquent query using orderBy() to display products in descending order of price.

PART-B (5 Marks Each)

1. Develop a Laravel form to accept **student name, email, and course** using the POST method. Implement CSRF protection and validate that all fields are required. Display appropriate validation error messages.
2. Explain **GET and POST form handling in Laravel**. Describe CSRF protection, retrieving old input, validation errors, and flash data.
3. Create a Laravel migration for an **employees** table containing id, name, email, salary, and department. Create an Employee model and use Eloquent ORM to **insert and retrieve employee records** from the database.
4. Create a Laravel application with a Country model and database table containing appropriate fields such as name, area, and population. Write an Eloquent query using orWhere() to retrieve countries whose area is greater than a specified value or whose population is greater than a specified value, and display the retrieved records.
5. Explain **form validation in Laravel**. Discuss built-in validation rules and custom validation messages with suitable examples.
6. Develop a Laravel application that creates a database relationship between Employee and Department models using a migration. Use Eloquent ORM with groupBy() and having() or a subquery to find and display the highest-paid employee(s) in each department.
7. Explain **Laravel database configuration and migrations**. Describe the role of the .env file and the steps involved in creating a database table using migrations.
8. Develop a Laravel application using a Customer model with an Orders relationship. Use Eloquent's whereDoesntHave() method to retrieve and display customers who have never placed an order.
9. Develop a Laravel application using an Employee model with a self-referencing manager_id relationship. Use Eloquent's belongsTo() relationship to identify and display employees whose salary is higher than their respective managers.
10. What are **Models and Eloquent ORM** in Laravel? Explain their purpose and describe the basic methods used for fetching, inserting, updating, and deleting database records.
11. Develop a Laravel application using a Person model to identify duplicate email addresses in the database. Use Eloquent's groupBy() and having() methods to find and display email addresses that occur more than once.
12. Develop a Laravel application to perform **CRUD operations** on a Product model using Eloquent

ORM. Implement operations to **create, display, update, and delete** product records.

PART-C (10 Mark Each)

1. Explain **Laravel database configuration, migrations, Models, and Eloquent ORM** in detail. Describe how Laravel connects to MySQL using the .env file, how migrations are used to create tables, and how Models and Eloquent are used to interact with database records.
2. Develop a Laravel application using the Eloquent ORM to retrieve movies from the movies table. Use where() clauses to display movies with an odd id and a description that is not equal to 'boring', and use orderBy() to arrange the results appropriately.
3. Mini project: build a validated Product CRUD app — migration + Eloquent model + a form using built-in validation rules with custom messages for insert/update, and an Eloquent listing query styled on the Module-4 filter tasks above.
4. Explain the complete **form processing and validation mechanism in Laravel**. Discuss GET and POST methods, CSRF protection, retrieving old input, validation errors, flash data, built-in validation rules, and custom validation messages.
5. Develop a Laravel application using a Weather model to compare each day's temperature with the previous day's temperature. Use Eloquent's whereColumn() or DB::raw() to retrieve and display the days when the temperature is higher than the previous day.
6. Explain **Eloquent ORM in Laravel** in detail. Discuss Models, Eloquent relationships, querying database records, where() clauses, orderBy(), groupBy(), having(), and other commonly used Eloquent methods with suitable examples.
7. Develop a Laravel application using an Employee model to find the second-highest salary from the database. Use Eloquent's orderBy() with skip() and take() or a suitable subquery to retrieve and display the employee with the second-highest salary.
8. Develop a Laravel application using Person and Address Eloquent models with a hasOne relationship. Use Eloquent ORM to define the relationship and fetch the person details along with the corresponding address without using raw SQL queries.
9. Explain **CRUD operations using Eloquent ORM in Laravel** in detail. Describe the process of creating, retrieving, updating, and deleting records, along with commonly used Eloquent methods for each operation.